

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 2

Adaptive Classified

OL-T44 · Circles, Arcs and Sectors

With Answers

Questions with full worked solutions

■ 3 pages

FAMILY MAP

Circles Arcs Sectors

F01

Recognise parts of a circle

1 Q | 4 marks

Original question 11

4MA1-1H Q10 Q10(a), Q10(b)

ORIGINAL QUESTION**4 marks**

- 10** Two circles, C_1 and C_2 , are drawn on a centimetre grid, with a scale of 1 cm for 1 unit on each axis.

The centre of circle C_1 is at the point with coordinates $(-1, 3)$ and the radius of C_1 is 13 cm.

The centre of circle C_2 is at the point with coordinates $(7, 18)$ and the radius of C_2 is 6 cm.

- (a) Work out the distance between the centre of C_1 and the centre of C_2

..... cm

(3)

- (b) Explain why circle C_1 intersects circle C_2

(1)

Worked solution 11

Chapter: Right-Angled Triangles - Pythagoras & Trigonometry; Circles, Arcs & Sectors

MODULAR UNIT 2

4 marks

Q10(a) – Chapter: Right-Angled Triangles - Pythagoras & Trigonometry

Family: Use Pythagoras in two dimensions • Variant: Find a hypotenuse

MethodThe coordinate changes between the centres are $7 - (-1) = 8$ and $18 - 3 = 15$.*Why: Subtract corresponding coordinates.***Method**Distance = $\sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289}$.*Why: Apply Pythagoras to the horizontal and vertical changes.***Method**

The distance between the centres is 17 cm.

*Why: Evaluate the square root and use the stated scale.***Final answer**

distance: 17 cm

Q10(b) – Chapter: Circles, Arcs & Sectors

Family: Recognise parts of a circle • Variant: Identify centre, radius, diameter, chord and circumference

MethodThe radii total $13 + 6 = 19$ cm, and the centre distance is 17 cm; since $17 < 19$, the circles overlap and intersect.*Why: Two circles can meet when the centre distance is less than the sum of their radii; here it is also greater than their radius difference.***Final answer****explanation:** They intersect because the centre distance 17 cm is less than $13 + 6 = 19$ cm